

TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
NATIONAL COLLEGE OF ENGINEERING
Examination Control Division
2080 Chaitra

Exam.	First Assessment		
Level	BE	Full Marks	60
Program	BCT	Pass Marks	24
Year / Part	III / I	Time	3 hrs

Subject: - Web Application Programming (ENCT 302)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt **All** questions.
- ✓ The figures in the margin indicate **Full Marks**.
- ✓ Assume suitable data if necessary.

1. What is the Internet and the Web? Explain the importance of Web technology and describe the concept of Web 2.0 with its key features. [2+2+4]
2. What is JavaScript? What are the potential platforms where JavaScript can be used? [2+4]
3. Explain JavaScript Execution Model. [6]
4. Explain JavaScript Modules with example. [4]
5. Explain MVC Architecture with neat diagram. [8]
6. Explain how to set, get and delete sessions. [4]
7. Define API (Application Programming Interface). Differentiate between Open APIs and Internal APIs. Explain why APIs are important in modern software development with purposes. [2+2+5]
8. Define Cross-Site Scripting (XSS). Differentiate between Stored XSS and Reflected XSS. Explain four potential impacts of XSS attacks on web applications. [2+2+4]
9. Write short notes on: [3+4]
 - a. Full-stack development
 - b. Frontend-backend integration

TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
NATIONAL COLLEGE OF ENGINEERING
Examination Control Division
2079 Chaitra

Exam.	First Assessment		
Level	BE	Full Marks	60
Program	BCT	Pass Marks	24
Year / Part	III / I	Time	3 hrs

Subject: - Web Application Programming (ENCT 302)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt **All** questions.
- ✓ The figures in the margin indicate **Full Marks**.
- ✓ Assume suitable data if necessary.

1. Explain the concept of multi-tier architecture in web development. Illustrate with a diagram a three-tier architecture consisting of: Presentation layer, Business logic layer, and Data layer. [3+5]
2. What is need for client side scripting? Explain features of JavaScript? [2+4]
3. What is JavaScript Event? Explain with Example. [2+4]
4. What are different data types in JavaScript. [4]
5. Explain MTV Architecture with neat diagram. [8]
6. Explain how to set, get and delete cookies from server side. [4]
7. What is REST? Explain the six architectural constraints of REST with their benefits. [2+7]
8. What is SQL Injection? Explain how SQL Injection attacks work with an example. Describe four preventive measures against SQL Injection in web applications. [2+2+4]
9. Write short notes on: [4+3]
 - a. Unit testing and the AAA pattern
 - b. Integration testing approaches

TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
NATIONAL COLLEGE OF ENGINEERING
Examination Control Division
2078 Chaitra

Exam.	First Assessment		
Level	BE	Full Marks	60
Program	BCT	Pass Marks	24
Year / Part	III / I	Time	3 hrs

Subject: - Web Application Programming (ENCT 302)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt **All** questions.
- ✓ The figures in the margin indicate **Full Marks**.
- ✓ Assume suitable data if necessary.

1. Describe the concept of the client/server computing paradigm and its role in modern computing. Illustrate with real-world examples of applications that utilize this architecture. [4+4]
2. What is NoScript? What are different way to include JavaScript in HTML document? [2+4]
3. What is Event Object? Explain with Example. [2+4]
4. How do you local storage in JavaScript. [4]
5. What is Django? Explain file structure of Django Project and Django App. [1+4]
6. Write a controller that accepts username and password as arguments and check with student table, if credential match, redirect to dashboard page otherwise display 'Invalid username/password'. Assume user already exists. [7]
7. What is the difference between PUT and PATCH HTTP methods? What status code is returned when a resource is successfully created using POST? Explain different HTTP status codes with their meanings including from success, from client error, and from server error categories. [2+2+5]
8. Define Cross-Site Request Forgery (CSRF). Explain how CSRF attacks exploit user sessions. Describe how Django's CSRF protection mechanism works with a code example. [2+3+3]
9. Write short notes on: [4+3]
 - a. Test-Driven Development (TDD)
 - b. Quality Assurance (QA) vs Testing

TRIBHUVAN UNIVERSITY
 INSTITUTE OF ENGINEERING
 NATIONAL COLLEGE OF ENGINEERING
 Examination Control Division
 2077 Chaitra

Exam.	First Assessment		
Level	BE	Full Marks	60
Program	BCT	Pass Marks	24
Year / Part	III / I	Time	3 hrs

Subject: - Web Application Programming (ENCT 302)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt **All** questions.
- ✓ The figures in the margin indicate **Full Marks**.
- ✓ Assume suitable data if necessary.

1. Describe the structure of an HTML document and explain different types of CSS. Write an HTML code snippet to demonstrate the basic structure with sectioning elements. [3+2+3]
2. What is NoScript? What is NaN in JavaScript? When does it occur? [2+2+2]
3. Create a form to input Name, Email, password. [10]
 - All fields are required.
 - Email should be valid.
 - Phone number should be valid. (9***** or 01*****)
 - Password must be at least 8 character long and it should contain at least one uppercase letter, one lowercase letter, one number and one symbol
4. What is callback hell? [4]
5. What is Routing? How do you handle routing in server side. Give examples. [1+4]
6. Write server side script to create and validate form with following rule and store given data into 'patients' table with details (name, patient_id, mobile, gender, address, dob, doctor name): [7]
 - Name, Mobile, Doctor Name, Gender, DOB: Required
 - Mobile: 10 digit start with 98, 97 or 96
 - DOB: YYYY-MM-DD format
7. What is statelessness in REST architecture? Explain the practices for Endpoints and URL design in RESTful APIs with good and bad examples for each. [3+6]
8. What is input validation? Differentiate between whitelist (allowlist) and blacklist (denylist) validation approaches. Explain why server-side validation is mandatory even when client-side validation is implemented. [2+3+3]
9. Write short notes on: [3+4]
 - a. Test automation benefits
 - b. Code review practices

TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
NATIONAL COLLEGE OF ENGINEERING
Examination Control Division
2076 Chaitra

Exam.	First Assessment		
Level	BE	Full Marks	60
Program	BCT	Pass Marks	24
Year / Part	III / I	Time	3 hrs

Subject: - Web Application Programming (ENCT 302)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt **All** questions.
- ✓ The figures in the margin indicate **Full Marks**.
- ✓ Assume suitable data if necessary.

1. Explain the rowspan and colspan attributes in HTML tables with examples. [4+4]
What are the attributes of list tags? Explain with suitable examples.
2. Explain the concept structure and importance of DOM. [6]
3. What are the major differences between Session storage and Cookie storage? [5]
4. Explain spread operator with example. [5]
5. What is Templating? How do you generate dynamic HTML content in server side. [1+4]
6. Design following forms and write corresponding server side code to store the user's values after satisfying following validation rules: [7]
 - Length of Full name up to 40 characters
 - Email address must be valid email address
 - Username must start with string and followed by number
 - Password length must be more than 8 characters
7. What is API versioning and why is it needed? List the three common approaches for API versioning. Explain API structure best practices including consistent response format with JSON examples for success and error responses, and pagination implementation. [2+3+4]
8. Define input sanitization. Explain the difference between input validation and input sanitization. Describe HTML encoding and its role in preventing XSS attacks. [2+3+3]
9. Write short notes on: [3+4]
 - a. Continuous Integration (CI)
 - b. Continuous Delivery vs Continuous Deployment

TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
NATIONAL COLLEGE OF ENGINEERING
Examination Control Division
2075 Chaitra

Exam.	First Assessment		
Level	BE	Full Marks	60
Program	BCT	Pass Marks	24
Year / Part	III / I	Time	3 hrs

Subject: - Web Application Programming (ENCT 302)

- ✓ *Candidates are required to give their answers in their own words as far as practicable.*
- ✓ *Attempt **All** questions.*
- ✓ *The figures in the margin indicate **Full Marks**.*
- ✓ *Assume suitable data if necessary.*

1. Create an HTML page containing an iframe within a paragraph (source: [4+4]
http://www.tuiost.edu.np, dimensions: 200px height, 400px width). Include tags for header, article, and footer. Insert a link containing mailto:info@iost.edu.np in the footer tag and set the keywords "iost", "csit" using Meta tag.
2. Explain different method that are available to access DOM element using id, [6]
class, name and selector with suitable example.
3. Write JavaScript code to change background color of web page every second. [6]
4. Explain Statement, Expression and Keywords in JavaScript. [4]
5. What are static files? What is ORM? What are benefits of ORM? [2+2+2]
6. Write a server side code to upload a file and validate: [6]
 - File extension (only allow image files: jpg, jpeg, png, gif)
 - File size (must be less than 2MB)
7. Define JSON (JavaScript Object Notation). List the syntax rules of JSON. [2+2+5]
Explain JSON data types with examples and provide a complete JSON example showing simple values, nested objects, arrays, and null values.
8. What are environment variables? Explain why sensitive information like [2+4+2]
database passwords and API keys should be stored in environment variables instead of source code. Demonstrate how to use environment variables in Django settings.
9. Write short notes on: [4+3]
 - a. CI/CD pipelines
 - b. GitHub Actions

TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
NATIONAL COLLEGE OF ENGINEERING
Examination Control Division
2074 Chaitra

Exam.	First Assessment		
Level	BE	Full Marks	60
Program	BCT	Pass Marks	24
Year / Part	III / I	Time	3 hrs

Subject: - Web Application Programming (ENCT 302)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt **All** questions.
- ✓ The figures in the margin indicate **Full Marks**.
- ✓ Assume suitable data if necessary.

1. Define CSS selectors. Explain different types of CSS selectors (element, class, id, attribute, universal) with examples. [1+3]
2. What are CSS combinators? Explain different types with suitable examples [1+3]
3. What are components in React. Demonstrate how props is passed to React components. [6]
4. Create a HTML page containing a division with image and its description in paragraph. Write a JavaScript function to change the value of description in the paragraph during onmouseover and onmouseout on the image. [6]
5. Explain Promise with example. [4]
6. What is Database Migration? What are the major differences between Sessions and Cookies? [2+4]
7. What are different field types for defining models in Django. Give example. [6]
8. Define XML (eXtensible Markup Language) and write syntax rules for XML. [3+6]
Explain XML structure components (Declaration, Element, Attribute, Comment, CDATA) with examples and provide a complete XML example showing nested elements, attributes, and arrays representation.
9. Explain the purpose of the following cookie security attributes: Secure, HttpOnly, and SameSite. Write the Django settings to configure secure session cookies. [6+2]
10. Write short notes on: [3+4]
 - a. Responsive web design
 - b. Mobile-first approach

TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
NATIONAL COLLEGE OF ENGINEERING
Examination Control Division
2073 Chaitra

Exam.	First Assessment		
Level	BE	Full Marks	60
Program	BCT	Pass Marks	24
Year / Part	III / I	Time	3 hrs

Subject: - Web Application Programming (ENCT 302)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt **All** questions.
- ✓ The figures in the margin indicate **Full Marks**.
- ✓ Assume suitable data if necessary.

1. What is the CSS box model? Explain margin, padding, border, and content with a neat diagram. Differentiate between block and inline elements with examples. [2+3+3]
2. Write a program which includes a function sum(). This function sum() should be designed to add an arbitrary list of parameters. (For e.g., if you call the function sum() as sum(1, 2) it should return the result 3 and if again you call the function sum() as sum(1, 3, 4) it should return the result 8). [6]
3. What is jQuery? What is the use of jQuery? How jQuery is differing from JavaScript? [6]
4. Explain principles of React. [4]
5. What is middleware? Create custom middleware for logging. [2+4]
6. Explain with example syntax of Django Template Language. [6]
7. How do you parse JSON data in Python using the 'json' module? How do you convert Python dictionary back to JSON string? Write a Python program to parse a JSON string containing book information (id, title, author, year, genres array), access and print each field, modify the data (add rating, append to genres), and convert back to JSON. [2+2+5]
8. Differentiate between authentication and authorization. Explain why storing passwords in plain text is dangerous. Describe the concept of password hashing and its key properties. [2+2+4]
9. Write short notes on: [4+3]
 - a. Web app manifest
 - b. Build pipeline in web development

TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
NATIONAL COLLEGE OF ENGINEERING
Examination Control Division
2072 Chaitra

Exam.	First Assessment		
Level	BE	Full Marks	60
Program	BCT	Pass Marks	24
Year / Part	III / I	Time	3 hrs

Subject: - Web Application Programming (ENCT 302)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt **All** questions.
- ✓ The figures in the margin indicate **Full Marks**.
- ✓ Assume suitable data if necessary.

1. Define media queries. Explain responsive web design using media queries with examples. [1+3]
2. What is Bootstrap? Explain the role of Bootstrap in responsive web design. [1+3]
3. What is arrow function? Write rules of writing JSX in React. [2+4]
4. How can you create a simple modal popup using HTML, CSS, and JavaScript that opens and closes on button clicks? [6]
5. List different types of jQuery selectors with one example of each. [4]
6. What is a database? What are types of databases? [2+4]
7. How do you inherit templates and how do you include partial template snippets. [6]
8. Define serialization in the context of Django REST Framework. Define deserialization and explain why it is important. Explain the complete serialization and deserialization flow in Django REST Framework for both incoming requests and outgoing responses with step-by-step process. [2+2+5]
9. Explain session-based authentication and its working mechanism. Describe three session security concerns (session hijacking, session fixation, and session prediction) and their prevention methods. [4+4]
10. Write short notes on: [4+3]
 - a. DevOps principles
 - b. DevOps lifecycle

TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
NATIONAL COLLEGE OF ENGINEERING
Examination Control Division
2071 Chaitra

Exam.	First Assessment		
Level	BE	Full Marks	60
Program	BCT	Pass Marks	24
Year / Part	III / I	Time	3 hrs

Subject: - Web Application Programming (ENCT 302)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt **All** questions.
- ✓ The figures in the margin indicate **Full Marks**.
- ✓ Assume suitable data if necessary.

1. Write an HTML script to insert an audio file in an HTML page using controls and autoplay properties. Also explain how to insert different multimedia elements in HTML. [4+4]
2. Write a program that displays the continuous time in the web page. The time should be in the format of HH:MM:SS. [2+4]
3. Explain AJAX. List benefits of using AJAX. Demonstrate AJAX with jQuery. [2+2+6]
4. Compare Django, Flask, FastAPI, ASP.NET, Spring Boot, and Node.js. [4]
5. Design following form in HTML and write corresponding server-side script to upload and submit user's data into database in consideration of following validation rules. [8]
 - **Form fields:** TU Registration Number, Email Address, Upload your Project File
 - **Validation rules:**
 - Registration number, email and upload file are mandatory field
 - Email address should be a proper email format
 - Upload file format must include pdf, doc, docx, ppt, pptx, jpeg file format
 - File size must be less than 5MB
6. Define microservices architecture. Define monolithic architecture. Compare monolithic and microservices architecture including key differences (deployment, scaling, database, technology), when to use each approach, and diagram representation of each. [2+2+5]
7. What is Cross-Origin Resource Sharing (CORS)? Differentiate between simple requests and preflight requests. Explain four CORS response headers and their purposes. [2+2+4]
8. Write short notes on: [4+3]
 - a. Progressive Web Apps (PWAs)
 - b. Breakpoints in responsive design
